

ADI Data Conversion Handbook 第五章

问题集锦

1. Page 271的Figure 5.11 要怎么理解？

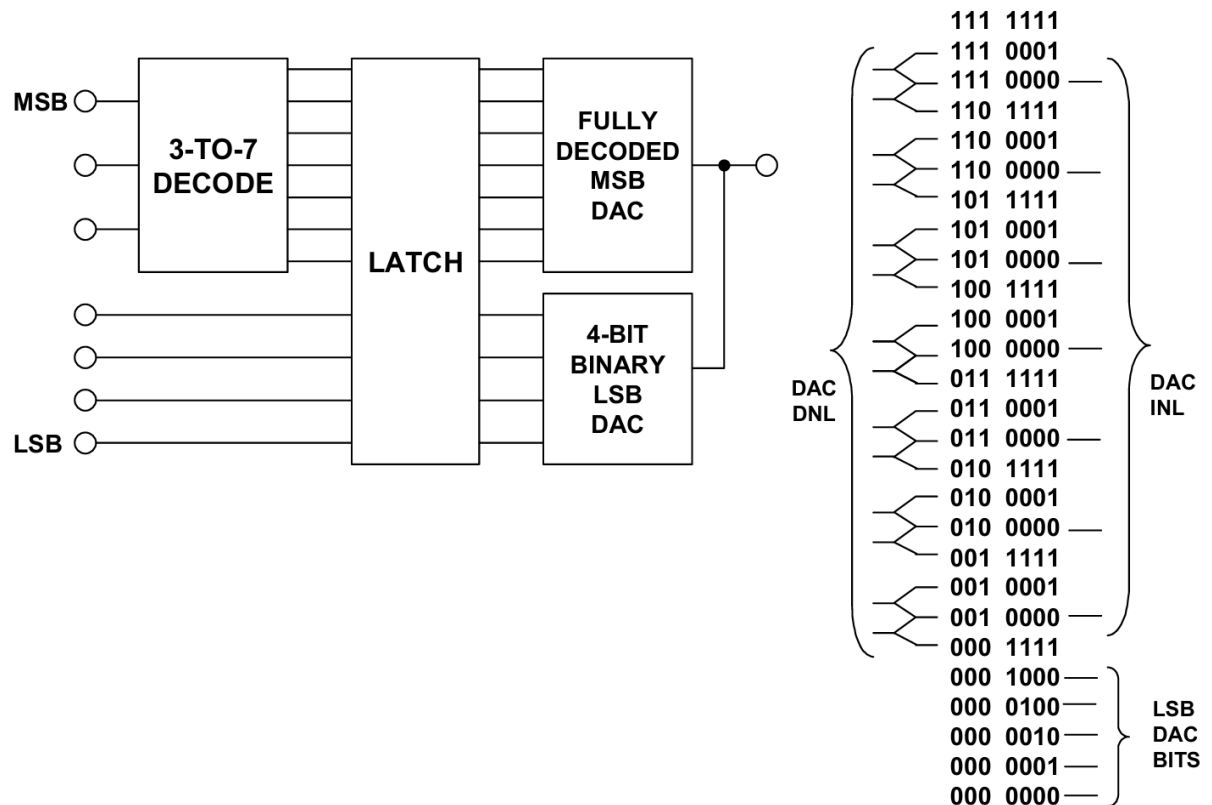


Figure 5.11: 3-Bit x 4-Bit Segmented 7-Bit DAC Test Codes

- 利用matlab建模一个如图5.11所示的3+4-bit的DAC并plot出传输特性曲线。
- 图5.12所示的settling error如何建模？
- 为什么图5.17中的SFDR会有两种不同的单位。
- Page 286, "Bipolar zero error in DACs using offset-type coding is a derived, rather than a fundamental quantity, because it is actually the sum of the bipolar offset error, the bipolar gain error, and the MSB linearity error." 应该如何理解这一句话？
- Page 286, "In addition, by using end-point linearity, we have defined the linearity error at zero and full-scale to be zero."
- Page 287, "This DAC has an error in the first and second bit weights. Bit 1 is 1.5 LSBs low, and Bit 2 is 1.5 LSBs high." 每个输入输出对应的DNL是怎么得到的？
- Page 292, 如何理解Settling Time这一部分的两种settling time的定义。
- Page 293, 什么是midscale settling time？
- Page 296: "The spectrum analyzer chosen to measure the distortion and noise performance of the DAC should have at least 10 dB more dynamic range than the DAC being tested." 为什么要大10dB？
- P296: "The analyzer resolution bandwidth must be set low enough so that the harmonic products can be resolved above the noise floor." 为什么要这么做？
- Page 298, 解释一下这段话: "The accuracy of this measurement should be verified by enabling the 9th bit of the DAC and ensuring that the analyzer noise floor drops by 6 dB."

If the noise floor does not drop by 6 dB, the measurement should be repeated using only the first 6 bits of the DAC. If near theoretical SNR is not achieved at the 6-bit level, the DAC under consideration is probably not suitable for ac applications where noise and distortion are important."

13. Page 300, 如何理解图5.23上这一段话。

In lieu of specific frequency measurements, there is another useful test method that gives a good overall indicator of the DAC performance at various combinations of output and clock frequencies. Specifically, this involves testing distortion for output frequencies, f_o , equal to $f_c/3$ and $f_c/4$. In practice, the output frequency is slightly offset by a small amount, Δf , where Δf is a noninteger fraction of f_c , i.e., $\Delta f = kf_c$, where $k \ll 1$. For an output frequency of $f_c/3 - \Delta f$, the even-order harmonics are spaced at intervals of Δf around the fundamental f_o output frequency as shown in Figure 5.23. The worst even-order harmonic is measured at various clock frequencies up to the maximum allowable while maintaining this same ratio. The same procedure should be repeated for an output frequency $f_c/4 - \Delta f$, in which case the odd-order harmonics are uniformly spaced around the output frequency as shown in Figure 5.24.

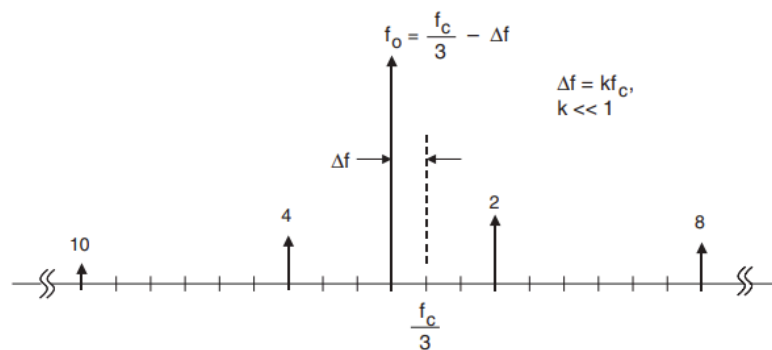


Figure 5.23: Location of Even Harmonics for $f_o = f_c/3 - \Delta f$

300

14. Page 305: "It should be noted, however, that this method only works well if the ADC (and the input voltage source) has an effective peak-to-peak input-referred noise that is less than 1 LSB. Larger amounts of input-referred noise tend to mask the transitions and make the measurement increasingly difficult. ". 这句话中的input-referred noise, 如果排除量化噪声, 这个我可以理解为INL的相关噪声部分吗?

15. Page 305: 图5.28以及P5.29最下面这句话: "Displaying the true and complement of each ADC bit makes determining the precise centers of the code transition points easier by simply adjusting the analog input voltage for equal brightness between the true and complementary LED display for the bits which are "fluttering." Additional logic is obviously required if the output of the ADC is in serial format." 这部分我不理解的有两点, 第一点是什么还要使用反码呢? 我觉得只看原码的变化就可以知道阶梯在哪里了。另外一个疑问是关于serial format 这个串行的意思是不是说从高位到低位依次变化呢? 最后, 我认为下图错了, 就算是顺次变化, 这个图我觉得也解释不通。

16. Page 305: "The advantage of using the code transition method directly is that the DNL for a particular code is simply the difference between the corresponding code transitions."

17. Page 306: "Although quite simple, the manual test setup is very useful with low noise ADCs, especially if only a few code transitions need to be measured, such as when

determining offset and gain errors. Figure 5.29 shows the details for measuring ADC gain and offset in terms of the code transitions. The example is for a 3-bit ADC, but is applicable to any resolution. " 这里我想问的是, 是不是测量offset只需要第一个代码转换点, 而gain error需要第一个和最后一个代码转换点? "

18. Page 306, "The subtraction function is implemented by complementing the ADC output code before driving the DAC, so that the DAC output is an inverted quantized representation of the ADC input. A simple resistive summer using a potentiometer can then be used to combine the signals." 如何实现的波形相减?
19. Page 306, Back-to-back static ADC testing: 这里我的疑问是将DAC减小频率(f_s/k)是为了等DAC的稳定输出, 但我是觉得ADC的频率高, 数据也就多, 那DAC降低频率的话, 有一部分数据就会丢失, 所以这个k的选取是不是前提要求不能丢失太多数据呢? "If desired, the DAC update clock can be a submultiple of the ADC sampling clock to ease the settling time requirements on the DAC and make the error waveform easier to observe."
20. Page 307: "A very low frequency linear ramp is applied to the ADC, and the potentiometer adjusted to null out the effects of gain differences between the ADC and the DAC." 关于调整误差, 这个是通过输出的INL曲线观察进行调整吗, 通过调整DC OFFSET以及电位器将两头都调整到0, 另外我不是很理解为什么 电位器能调整gain
21. Page 307 back-to-back static ADC testing部分要求DAC的性能高于ADC. 这里我想问的是, 这个要求是DAC的resolution、static、dynamic都强于被测试的ADC吗?
22. Page 307 back-to-back static ADC testing: 为什么测DNL时需要用"a low amplitude triangle waveform with a precision dc offset adjustment", 而测试INL时需要用"a full-scale triangle waveform".
23. Page 308: As the ADC resolution is increased, the frequency of the input signal must be made lower, the amplitude of the error waveform decreases, and the effects of ADC noise and DAC errors become more pronounced. 这里我不是很懂几句话的因果关系。
24. Page 312: 测试adc为什么输入信号要overdrive?
25. Page 313: "However, a nonmonotonic ADC will also generally have a higher level of distortion, and this condition is easily detected with an FFT analysis of the output data"
26. Page 316: 推导"Eq.5.20".
27. Page 317: 解释"deglitcher".
28. Page 317: "The vast majority of integrated circuit ADCs today are sampling ADCs that contain an internal sample-and-hold function of one form or another." Is there any ADC that is not sampling ADC?
29. Page 318: " The output low-pass filter is chosen to have a cut-off frequency of approximately $f_s/2.2 \cdot K$ so that images are attenuated over the bandwidth of interest. "
30. Page 319: "In practice, this reduces the effects of DAC settling time glitches, and the low frequency Δf signal can be easily observed on an oscilloscope for nonlinearities and missing codes, or the distortion measured with a spectrum analyzer. A Δf frequency of a few hundred kilohertz generally performs satisfactorily. Both the sampling clock and the input signal should be derived from stable frequency synthesizers or crystals to prevent phase noise on the low frequency Δf signal. "
31. P320: "The input sinewave is set to a low frequency and adjusted until the DAC indicates the ADC is barely clipping the positive and negative peaks of the input sinewave. The input frequency can then be increased, and the measurement repeated. An increase in

input signal level required to cause clipping corresponds to a decrease in the ADC gain; a decrease in input level implies an increase in gain. This test can be extended well beyond the Nyquist frequency, because the DAC response is not critical—it is only being used to detect the point of ADC clipping. Small-signal bandwidth can also be measured, where the input sinewave level is set to activate the same predetermined number of DAC output levels at each test frequency." 如何理解这一段话？

32. Page 320: "In summary, the back-to-back ADC test method can serve as a quick check of overall ADC ac performance for resolutions of up to about 10 bits. It is also useful in testing end-to-end performance in systems that use a reconstruction DAC in conjunction with an ADC. " 里的 end-to-end 是什么意思？
33. Page 321: 推导“Eq.5.20”。
34. 关于 Eq. 5.21, 以下说法是否正确: "\$Q_T\$是基于各电平等概率出现的前提, \$Q_A\$是基于实际sin函数电平出现的概率。我认为\$Q_A\$是实际量化误差(拟合正弦波得到的, 他的概率不是bathtub吗? 因此量化误差应有所不同才对), 而\$Q_t\$是理想的量化误差(由三角波输入得到的)"
35. Page 322, 如何理解公式5.23.
36. Page 322, 请解释: "Digital filters can be constructed based on their impulse response, because the coefficients of an FIR filter and its impulse response are identical. "
37. Page 324: "Although “complex” ADCs are not available, it is very common to use two ADCs to synchronously sample the I and Q data streams from a quadrature demodulator, in which the FFT input data is complex—however, this is a special case."
38. Page 332: "In order to prevent repetitive data patterns in the data record and ensure a random quantization noise spectrum, MC should be a prime number. " 有repetitive patterns会对哪些方面造成影响？
39. Page 333: "The default for the harmonic leakage is ± 3 bins, and the dc leakage default is 6 bins. All the leakage values can be modified if required, although the default values work well in most cases. " 什么情况下需要修改default value?
40. Page 335: why should we use a 6dB attenuator in "The 6 dB attenuator isolates the DUT (ADC) from the output of the filter"
41. Page 336: in figure 5.71, why we choose resistor values like this?
42. Page 336: 解释一下图5.72里提到的input overdrive是什么？
43. Page 337: "The residual signal inside the notch after passing through the transmission system represents clipping noise, thermal noise, and IMD distortion products." clipping noise指什么？
44. Page 338: 图5.76, 什么是"locked-histogram"? 为什么要用同一个low-jitter clock generator产生两路信号? 这两路信号之间有什么区别? 为什么时钟jitter同adc内部的jitter无法区分？
45. Page 343: flat强调什么？
46. 什么是"overvoltage recovery time"?
47. 解释图5.33里的Crossplot measurements of adc linearity。
48. 验证公式5.30.
49. 测试adc的inl/dnl之前要先做哪些准备工作？
50. 如何产生一个low distortion的sine wave inputs。

51. 假设ADC静态测试使用斜坡信号，那么要达到0.01LSB的精度要求，对斜坡信号有什么要求？
52. 为了保证静态测试的静态性，斜坡信号的update周期有什么要求？
53. 使用histogram+sin波测试inl/dnl的原理是什么？为什么要用sin波而不用ramp？
54. 承接上一题，既然sin波可以同时用于动态和静态测试，那么用于动态和静态测试的正弦波在频率上有何不同？
55. 已知采样频率，fft测试的点数，如何确定输入信号的频率？
56. Erbw的定义是什么？为什么adc的性能会随着频率的上升而下降？
57. 如何通过ADC输入短接时的噪声的 RMS 值来计算 ADC的ENOB。
58. 介绍一下DDS。